

Supplementary Material

PoxHostAtlas: A reproducible cross-study atlas of host gene-expression remodeling during poxvirus infection — Mubarak & Nguyen

All tables and figures are regenerated by the released pipeline. Raw data: GEO GSE278320, GSE284044, GSE287860, GSE288000, GSE185520.

Table S1. Complete dataset screening record (all 12 candidates identified by the systematic search).

Accession	Tier	Virus / host	n	Used for	Reason
GSE278320	A	Vaccinia (WR) / human HAP1	38	Primary meta	Verified count files; clean mock vs VacV
GSE284044	A	Vaccinia / Vero (time course)	24	Primary meta	Author DESeq2 tables; ortholog-mapped; temporal
GSE287860	A	Myxoma M003 effector / human	6	Primary meta	Direct count matrix; clean effector contrast
GSE288000	A	Myxoma M003 / human (5 KO)	30	Primary meta	Five-background robustness panel
GSE185520	B	Vaccinia D10 / human	30	Validation	Only FPKM/log2FC tables available
GSE288433	C	Myxoma MYXV screen / human	19	Excluded	Screen design needs barcode decoding
GSE329296	C	Poxvirus MURV-seq leaders	12	Excluded	5' leader assay; not host-DGE comparable
GSE228963	C	Vaccinia	6	Excluded	Viral transcripts only; no host control
GSE137757	C	Oncolytic MG1/VacV	16	Excluded	Mouse; TE-normalised; ortholog caveats
GSE280044	C	Vaccinia poxins (B2R)	6	Excluded	Single viral gene, not infection
GSE228918	C	Entomopoxvirus	81	Excluded	Insect hosts; no ortholog comparability
GSE259380	C	Oncolytic vaccinia TG6050	149	Excluded	Tumour-microenvironment context

Table S2. Host-program over-representation, direction-stratified.

Program	Set	Size	Hits	Expected	Fold	FDR
Cell cycle / mitosis	all	21	13	2.29	5.68	2.5e-07
Inflammatory / NF-κB	all	22	13	2.40	5.43	2.8e-07
RNA helicases (DEAD/DEAH)	all	56	8	6.10	1.31	0.87
Translation initiation (eIF)	all	70	7	7.62	0.92	1.0
Ribosomal proteins	all	189	14	20.58	0.68	1.0
Interferon-stimulated genes	all	40	1	4.36	0.23	1.0
Inflammatory / NF-κB	up	22	13	1.14	11.43	5.6e-11
RNA helicases (DEAD/DEAH)	up	56	8	2.90	2.76	0.039
Translation initiation (eIF)	up	70	5	3.62	1.38	0.98
Cell cycle / mitosis	down	21	13	1.20	10.83	8.6e-11
Ribosomal proteins	down	189	14	10.81	1.30	0.97
Translation initiation (eIF)	down	70	2	4.00	0.50	1.0

Direction-stratified enrichment resolves the programs more sharply than the pooled test. DEAD/DEAH helicases are not enriched overall (1.31, FDR 0.87) but are enriched among up-regulated genes (2.76, FDR 0.039).

Table S3. Program direction: pooled versus vaccinia-only contrasts.

Program	Genes (pooled)	Median log2FC	Genes (VacV)	Median log2FC (VacV)
Ribosomal proteins (cytoplasmic)	189	-0.327	84	-0.787
Cell cycle / mitosis	21	-0.801	21	-0.584
Translation initiation (eIF)	70	+0.186	49	-0.324
Mitochondrial ribosome	82	+0.111	76	-0.551
Cholesterol / sterol biosynthesis	18	-0.248	18	-0.704
Integrated stress response / UPR	21	+0.331	20	-0.321
Apoptosis / cell death	17	+0.135	15	-0.156
RNA helicases (DEAD/DEAH-box)	56	+0.319	47	+0.095
Interferon-stimulated genes (ISG)	40	+0.328	25	+0.214
Inflammatory / NF-κB signalling	22	+2.000	16	+0.744

Table S4. Reproducibility classification and leave-one-study-out stability.

Metric	Value
Class I — conserved, low heterogeneity	33
Class II — conserved, high heterogeneity	254
Class III — context-dependent	198
Total classified	485
Direction preserved in every leave-one-study-out re-pooling	327 / 485 (67.4%)
Significance preserved in every leave-one-study-out re-pooling	28 / 485 (5.8%)
Significance retained, GSE278320 removed	178 / 485 (36.7%)
Significance retained, GSE284044 removed	122 / 485 (25.2%)
Significance retained, GSE287860 removed	85 / 485 (17.5%)
Significance retained, GSE288000 removed	58 / 485 (12.0%)
DHX15 — direction and significance preserved in all four re-poolings	Yes

With four Tier-A studies, removing one leaves $k = 3$, so directional preservation is the informative stability measure at this sample size.

Table S5. Top 15 reproducibly up- and down-regulated host genes.

Up	k	log2FC	P	meta-FDR	Down	k	log2FC	P	meta-FDR
SLC1A2	3	+6.29	40	2.8e-03	BMP6	3	-3.67	98	5.5e-03
FCRL5	3	+6.11	82	1.2e-02	IGF1	3	-3.62	72	1.5e-02
IL6	3	+5.13	79	1.2e-79	SPTLC3	3	-3.49	94	6.3e-03
IL1B	3	+4.94	96	2.5e-15	DEPTOR	4	-3.42	87	1.5e-04
EDN1	3	+4.75	93	3.8e-04	CFD	3	-3.37	97	2.0e-02
ADORA2A-AS1	3	+4.49	94	1.3e-04	CDA	3	-3.33	54	2.1e-05
ASTL	3	+4.42	63	3.0e-02	CCDC33	3	-3.04	39	3.7e-02
GRIA4	4	+4.33	57	5.3e-04	C16ORF74	3	-3.01	29	1.4e-14
BIRC3	4	+4.23	93	1.5e-03	MEGF6	3	-2.90	98	1.0e-02
MAB21L1	3	+4.19	68	4.2e-04	RUFY4	3	-2.86	71	4.6e-05
FFAR2	3	+4.19	59	6.9e-78	TRIM50	3	-2.82	0	1.9e-04
SFRP2	3	+4.17	11	1.9e-02	LINGO3	3	-2.73	78	6.3e-10
SLAMF7	3	+4.15	99	4.2e-02	B3GNT7	4	-2.70	59	5.9e-03
MORC1	3	+4.12	3	4.7e-08	CDH5	3	-2.65	17	3.5e-05
TNFSF18	3	+4.00	29	4.4e-27	ACKR3	4	-2.55	58	1.2e-10

Table S6. Machine-learning controls (leave-one-dataset-out, whole studies held out).

Control / feature set	Balanced accuracy
Conserved meta-signature (29 genes)	0.794
Label permutation (mean of 1,000)	0.538
Housekeeping genes	0.822
Size-matched random gene sets (mean of 1,000)	0.742
Interferon-stimulated genes only	0.711
Study-identity prediction (3-fold CV accuracy)	0.933
Signature vs random-set null — percentile	81.7th
Signature vs random-set null — empirical p	0.18
Top-feature ablation vs random ablation	p = 0.003 (99.8th pct)

Controls reported in full. Label permutation collapses to chance and top-feature ablation degrades the model significantly. Housekeeping genes and size-matched random sets perform comparably to the signature, consistent with infection remodelling a broad host program rather than a narrow gene family; no specificity claim rests on these controls.

Table S7. Key analysis parameters.

Step	Parameter
Differential expression	PyDESeq2 on low-count-filtered counts; infected vs control
Ortholog mapping	Vero → human, GENCODE v44; 18,785 genes carried Vero-derived values
Meta-analysis	DerSimonian–Laird random effects, genes in ≥ 2 studies; I^2 and τ^2 reported
Multiple testing	Benjamini–Hochberg meta-FDR
Reproducibility threshold	≥ 3 studies, concordant direction, meta-FDR < 0.05, $ \log_2\text{FC} > 0.5$
Enrichment	Hypergeometric vs background of genes measured in ≥ 3 studies
Co-expression network	$ \text{Pearson } r \geq 0.7$ on study-centred 45-sample matrix; communities on positive edges
Classification	Elastic net, random forest, gradient boosting, linear SVM; leave-one-dataset-out
Code and data	Full pipeline, result tables and figures released with SHA-256 manifest and version lock

Figure S1. Full analysis pipeline, from dataset discovery to validation.

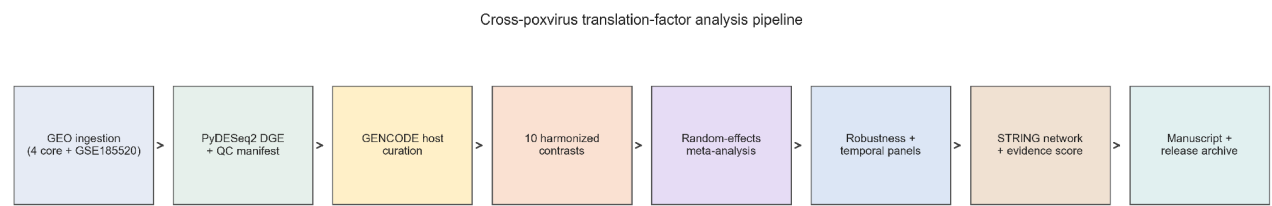
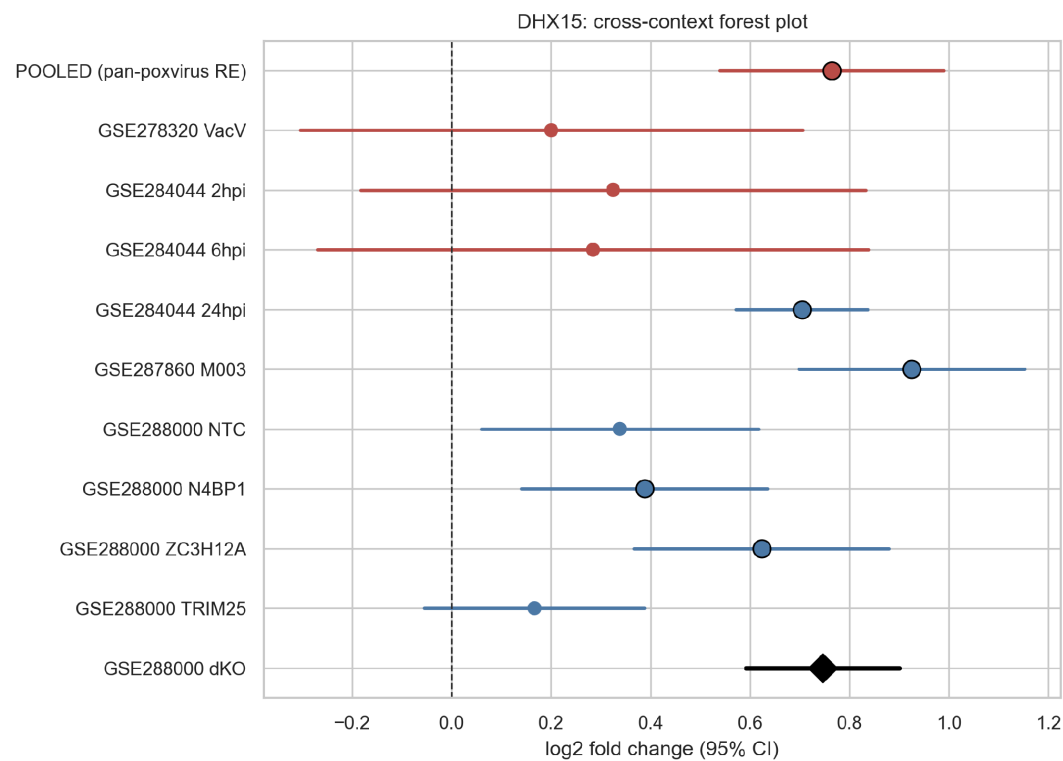


Figure S2. DHX15 cross-context forest plot (full size).



Shown at full size; the main-text Figure 5 presents this panel compressed alongside DHX29.

Figure S3. DHX29 cross-context forest plot (full size).

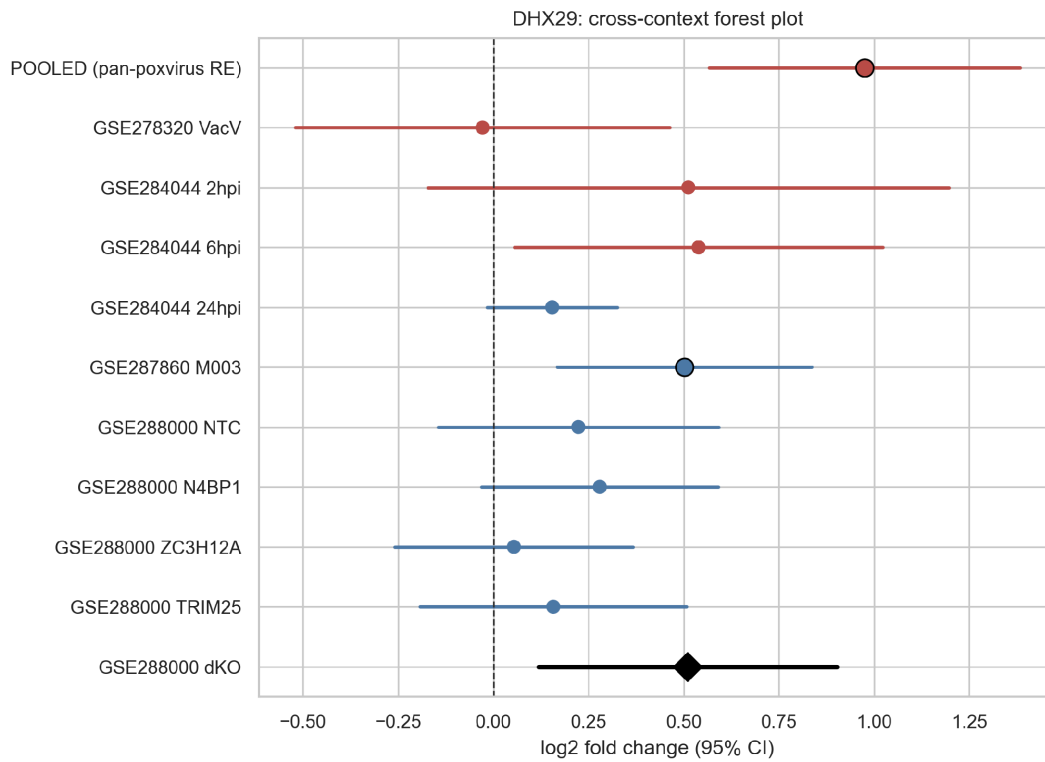
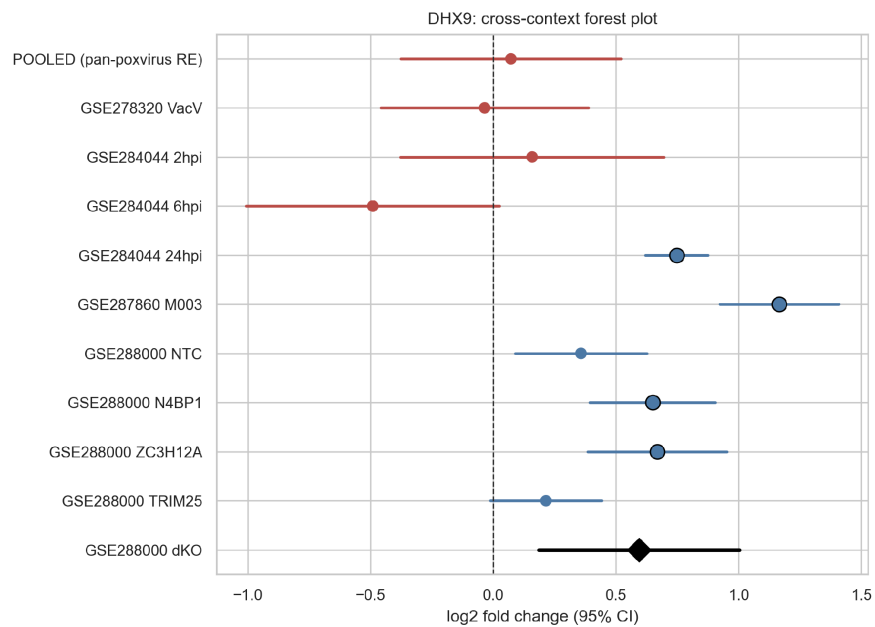


Figure S4. Additional helicase forests: DHX9, DDX21 and EIF4B.



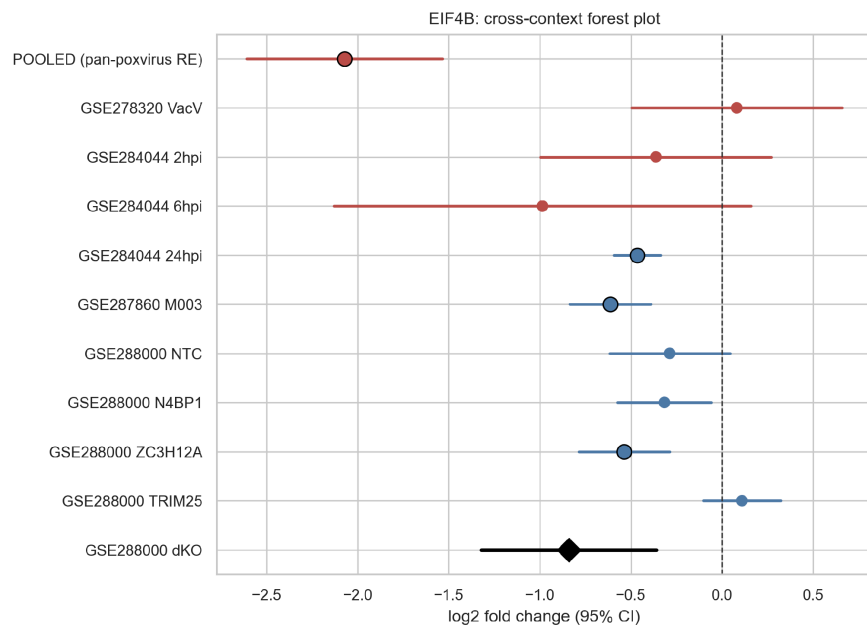
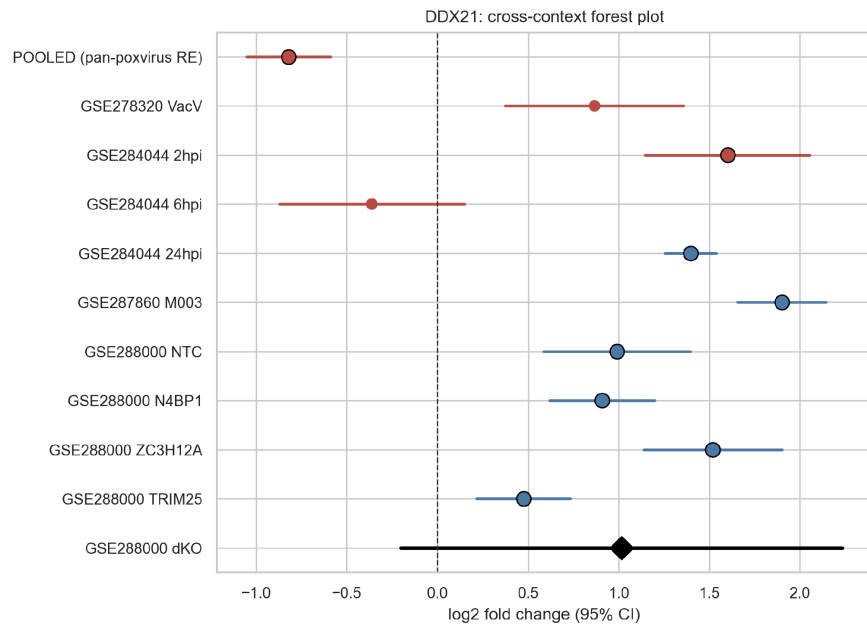
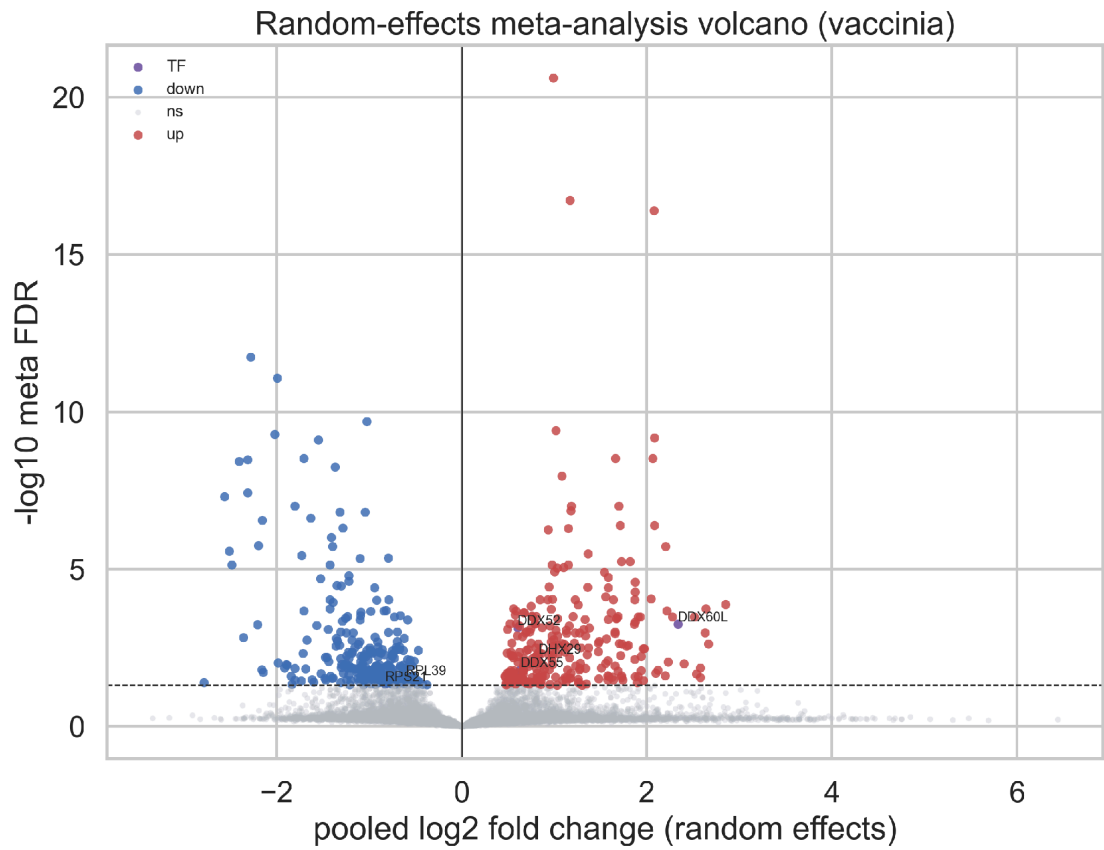


Figure S5. Vaccinia-only meta-analysis volcano.



Sensitivity layer isolating true infection from Myxoma effector-manipulation contrasts.

Figure S6. Vaccinia temporal dynamics across the 2, 6 and 24 h post-infection time course.

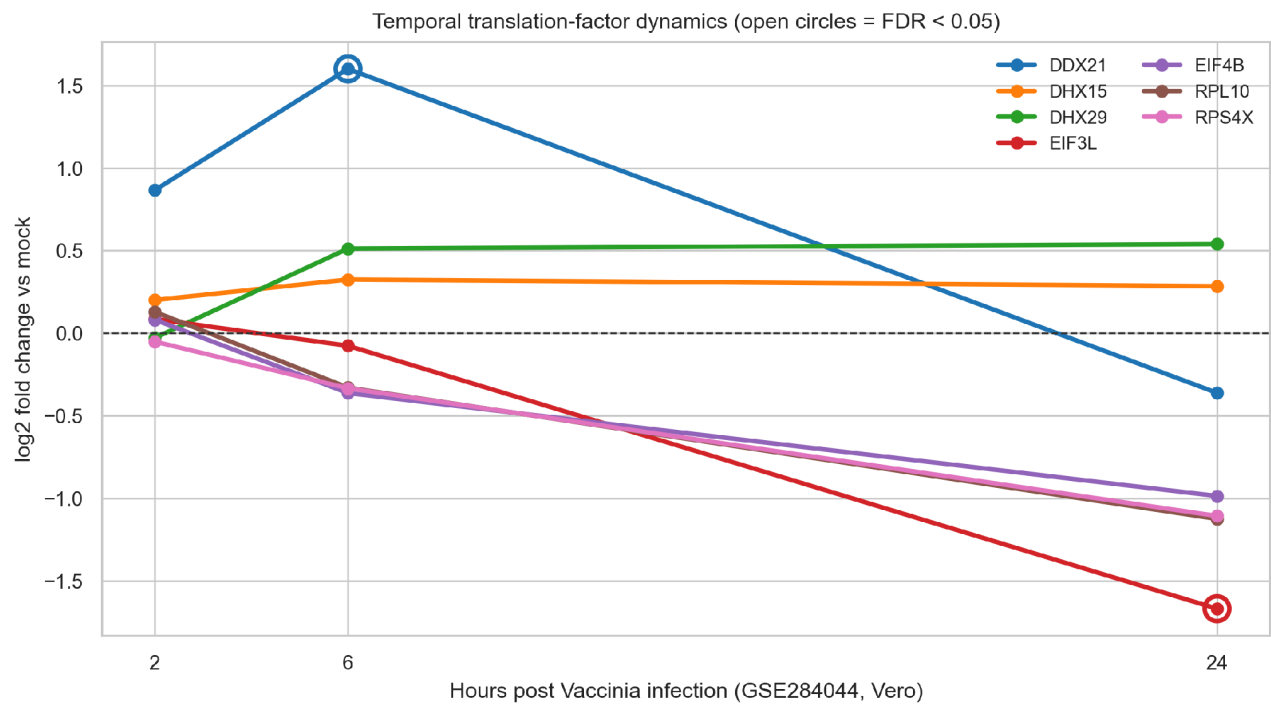


Figure S7. Host-background robustness across the five knockout backgrounds of GSE288000.

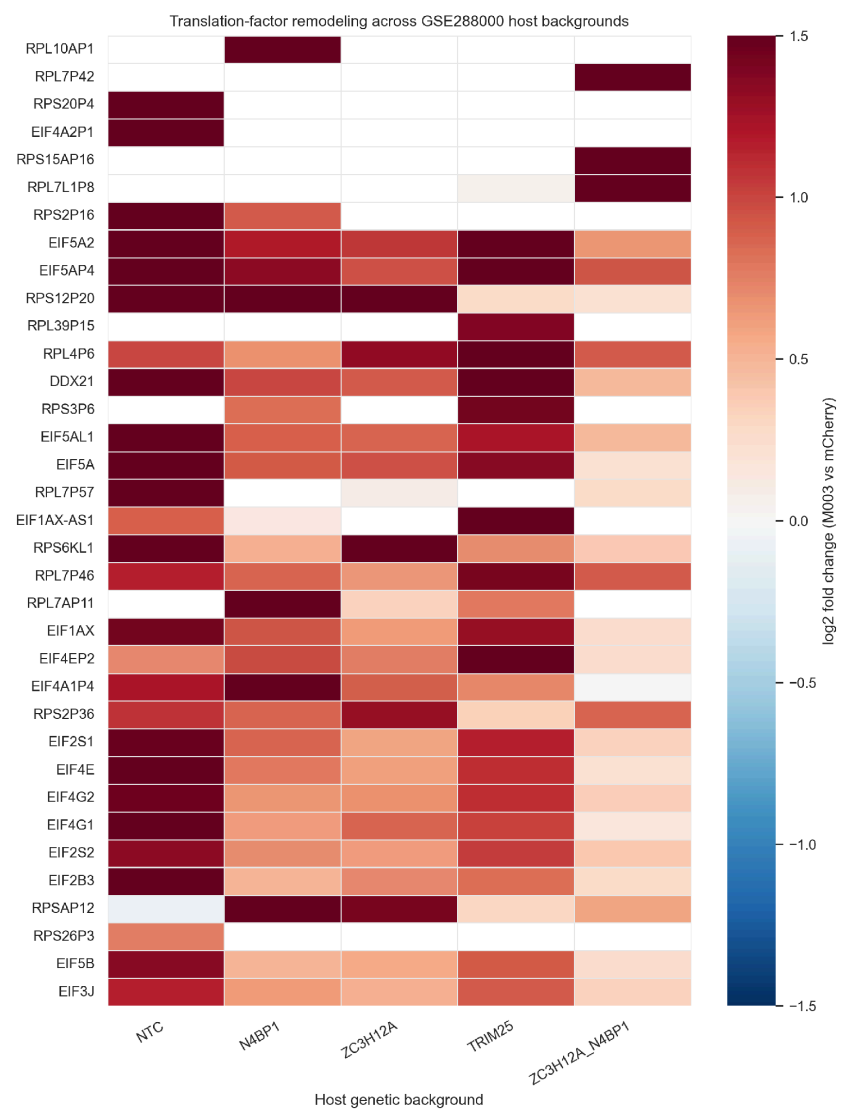


Figure S8. Cross-study rank correlation between Tier-A datasets.

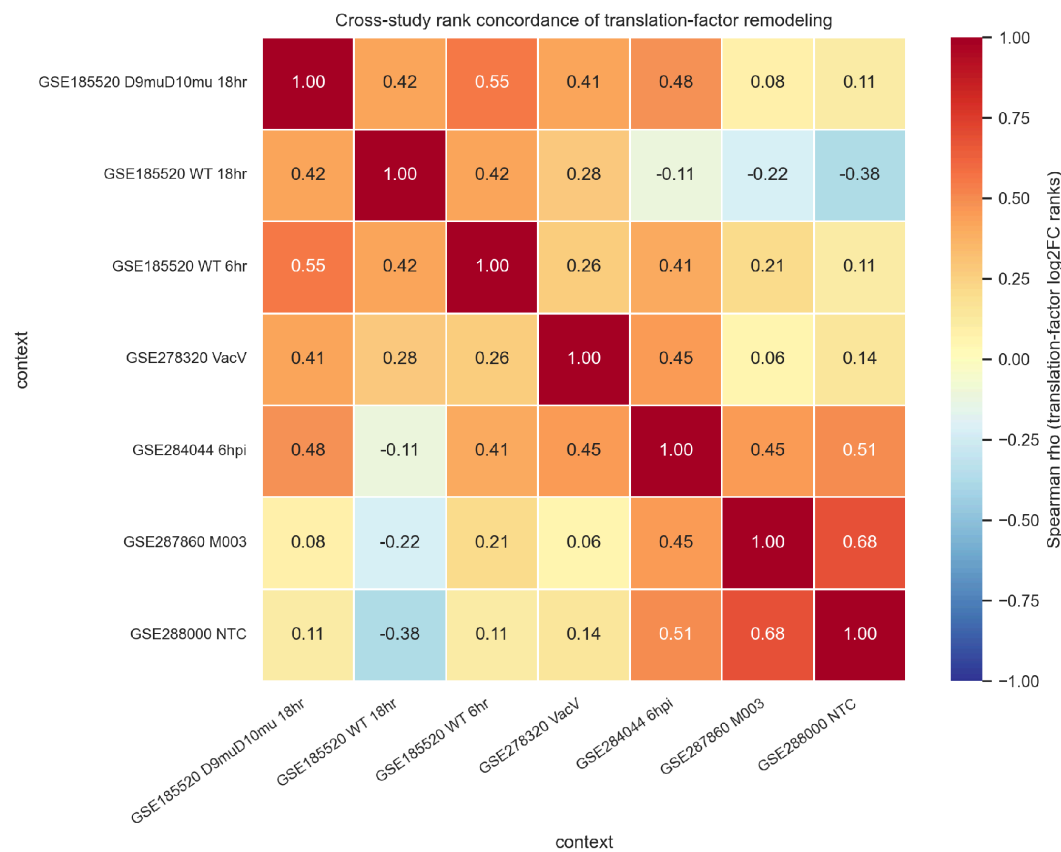
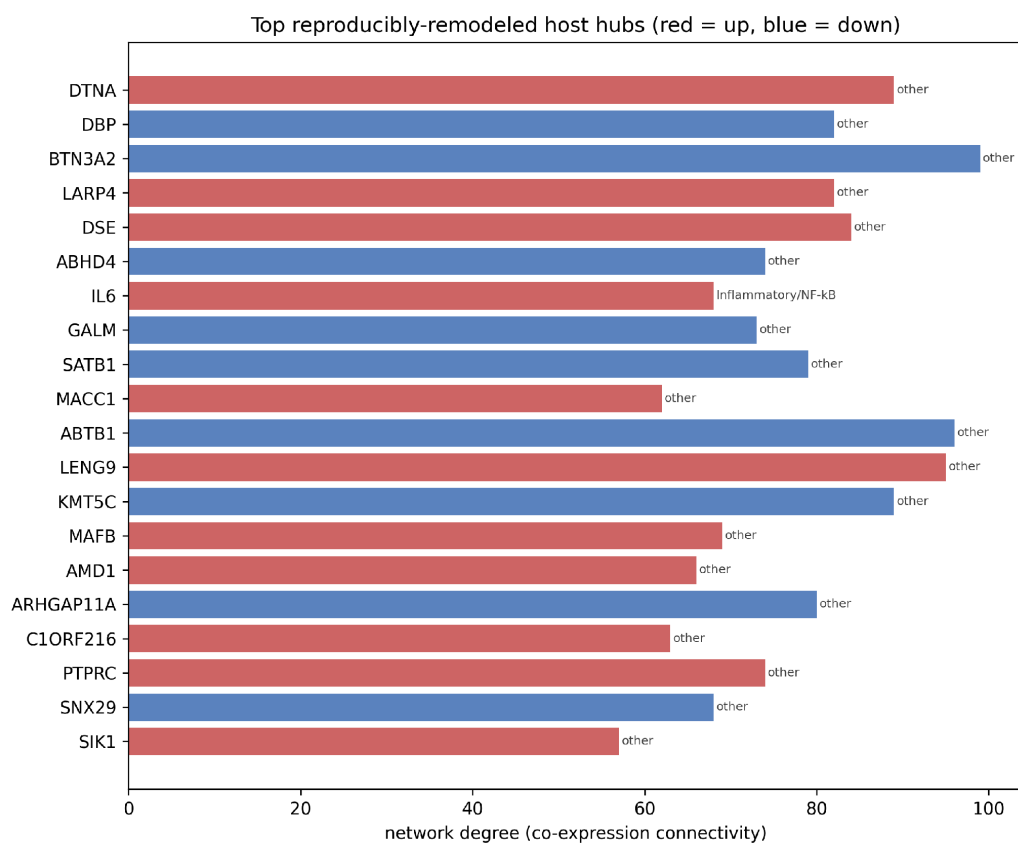


Figure S9. Ranked host hubs from the genome-wide co-expression network.



Visual companion to main-text Table 2.

Figure S10. STRING physical-interaction priority network for translation and helicase genes.

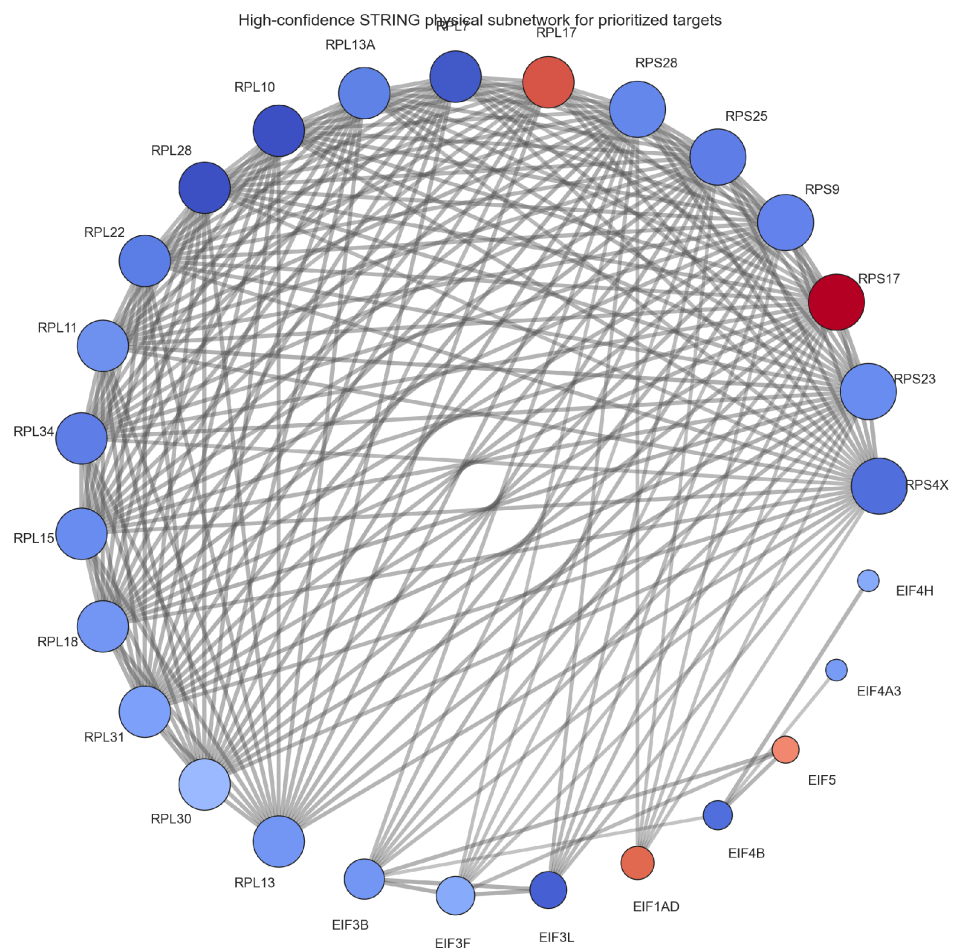


Figure S11. Candidate evidence matrix across meta-analysis, network and validation layers.

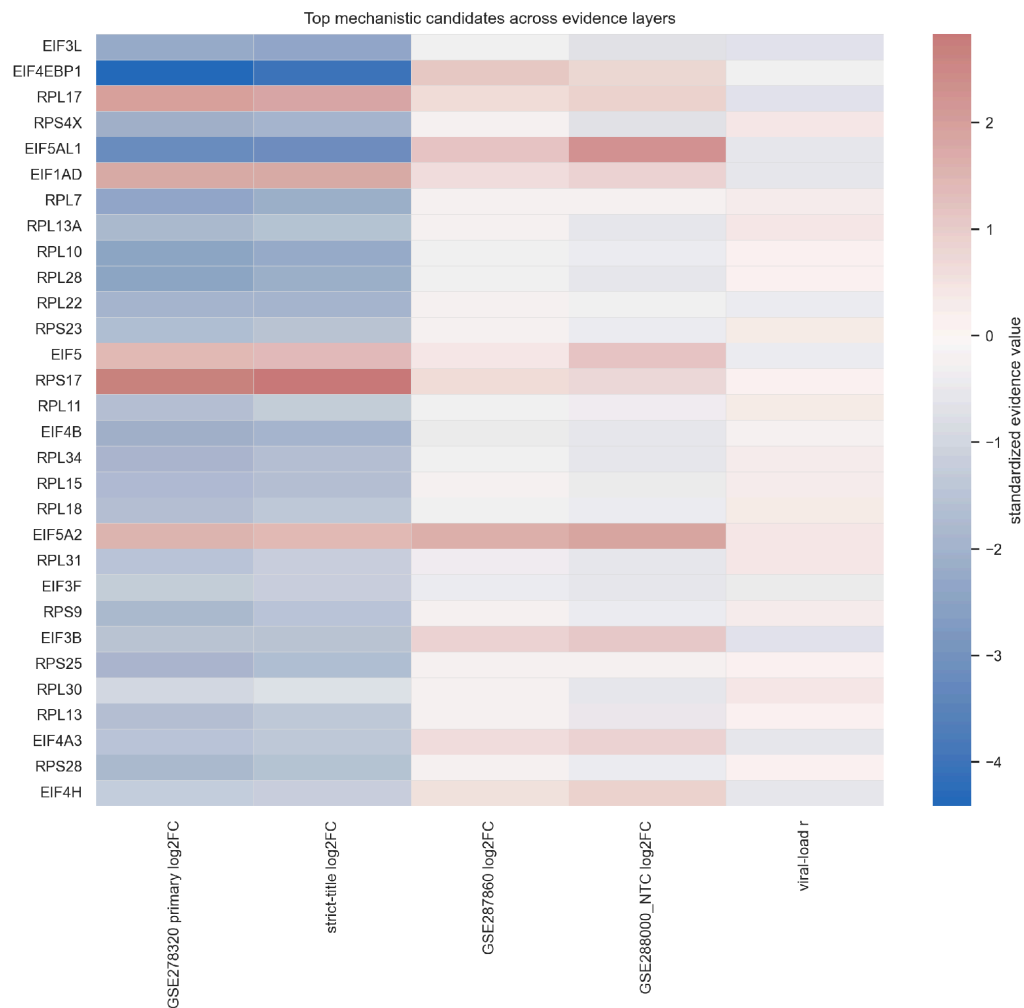


Figure S12. DEAD/DEAH-box helicase priority ranking.

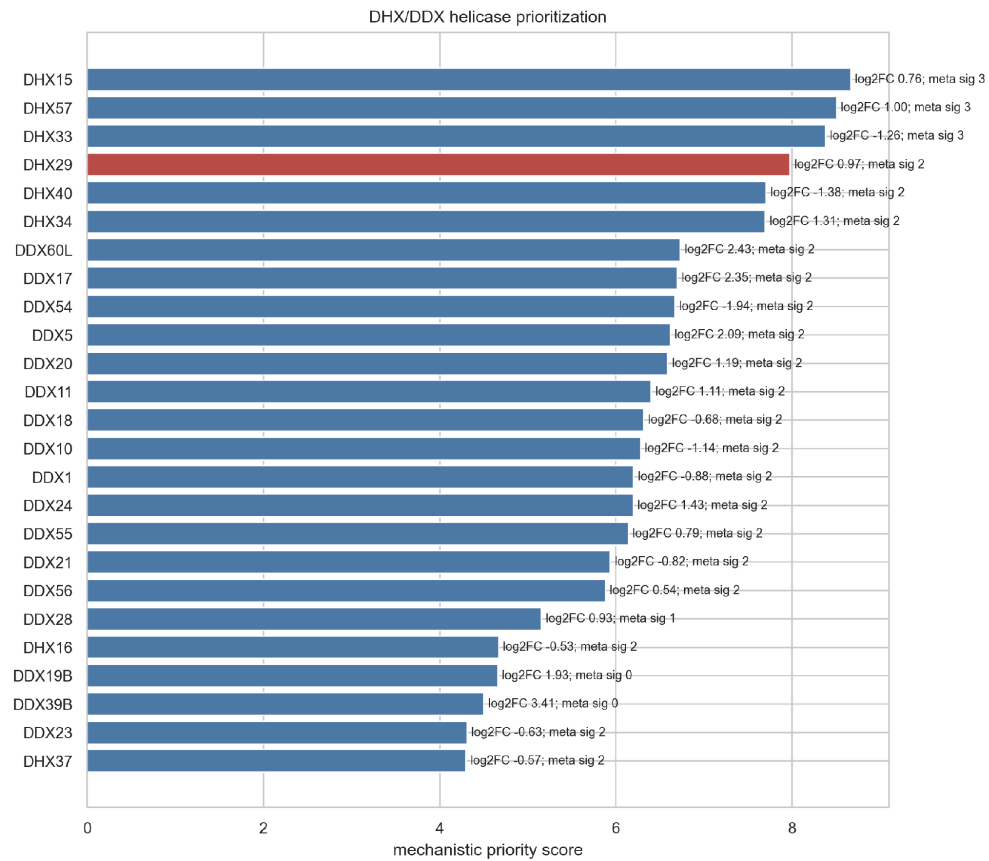


Figure S13. Extended evidence heatmap for top-ranked host genes.

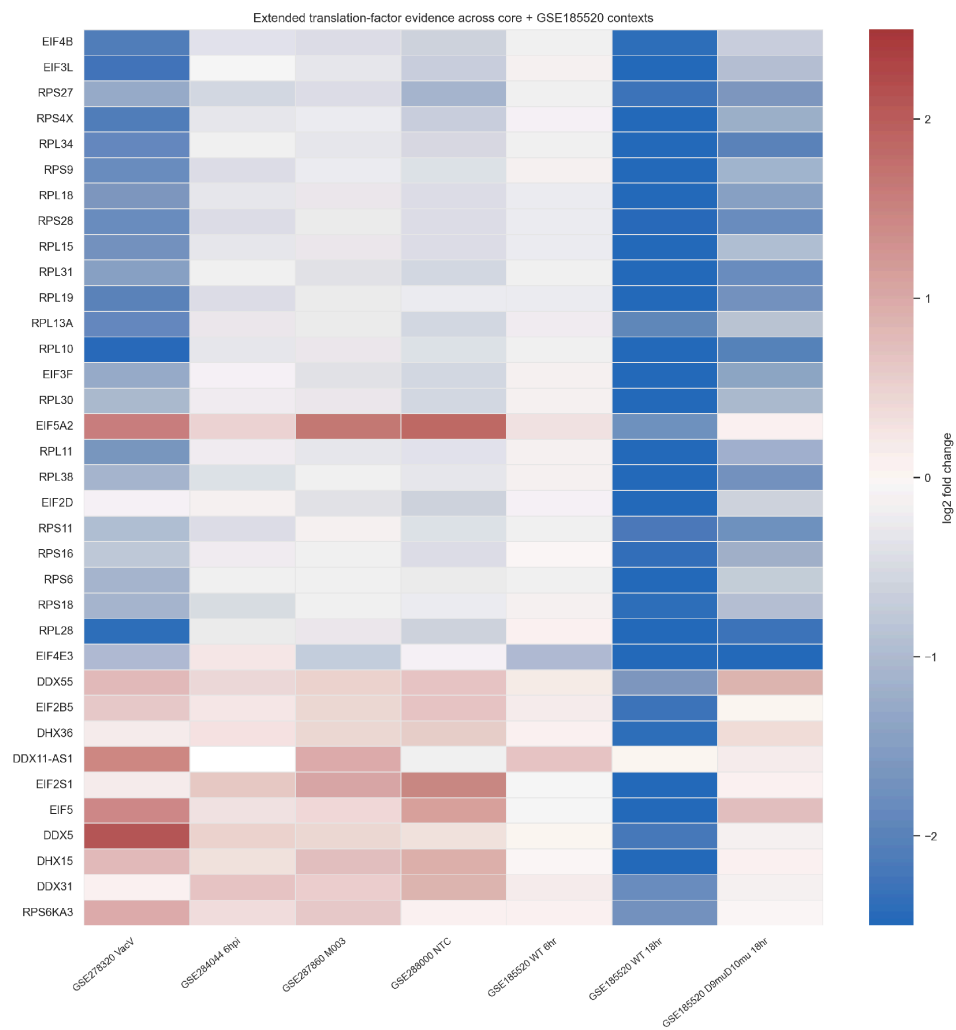


Figure S14. Independent single-cell validation: host shutoff in two cell types (Matia 2024).

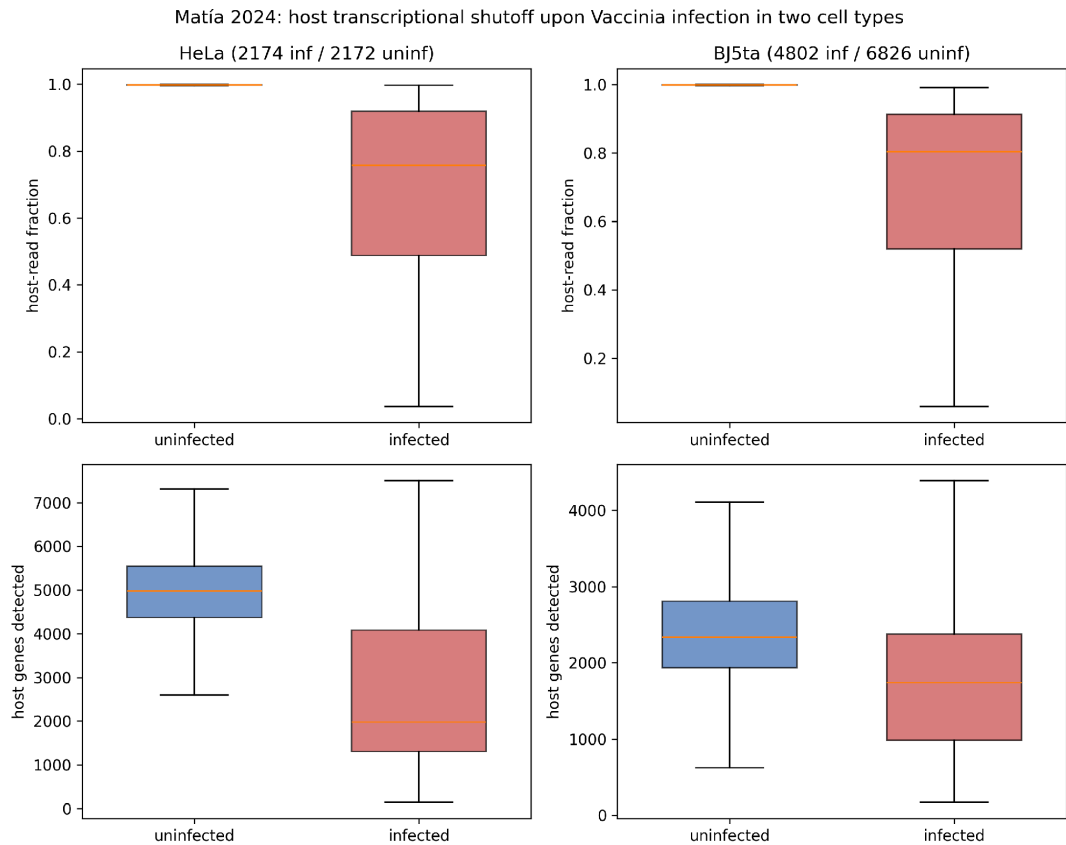


Figure S15. Cross-modality concordance between transcript and protein level.

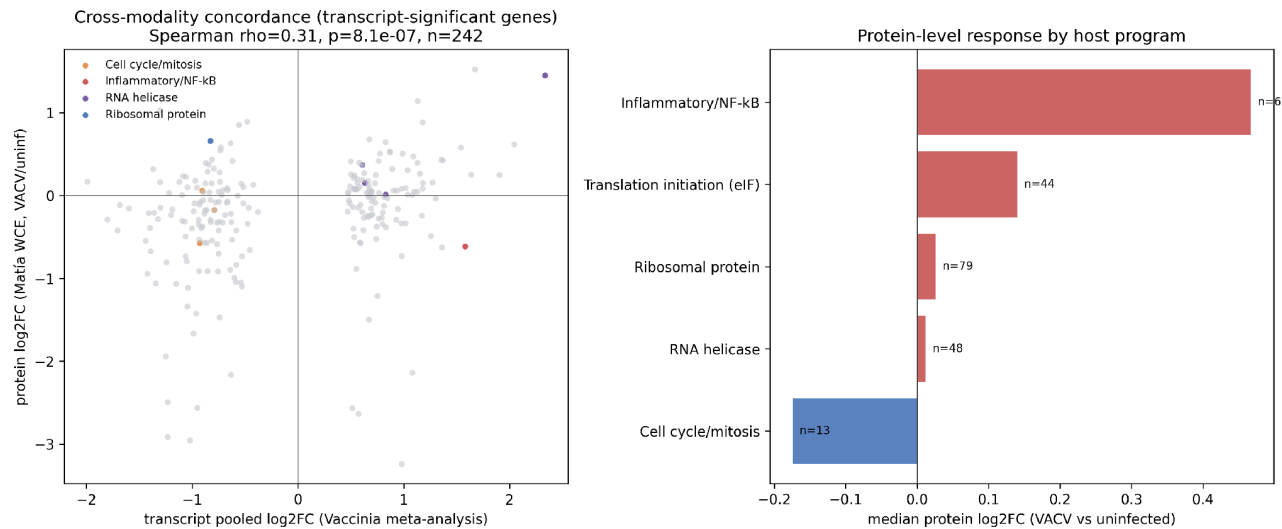


Figure S16. Module concordance summary across host programs.

Translation-module meta-analysis synthesis

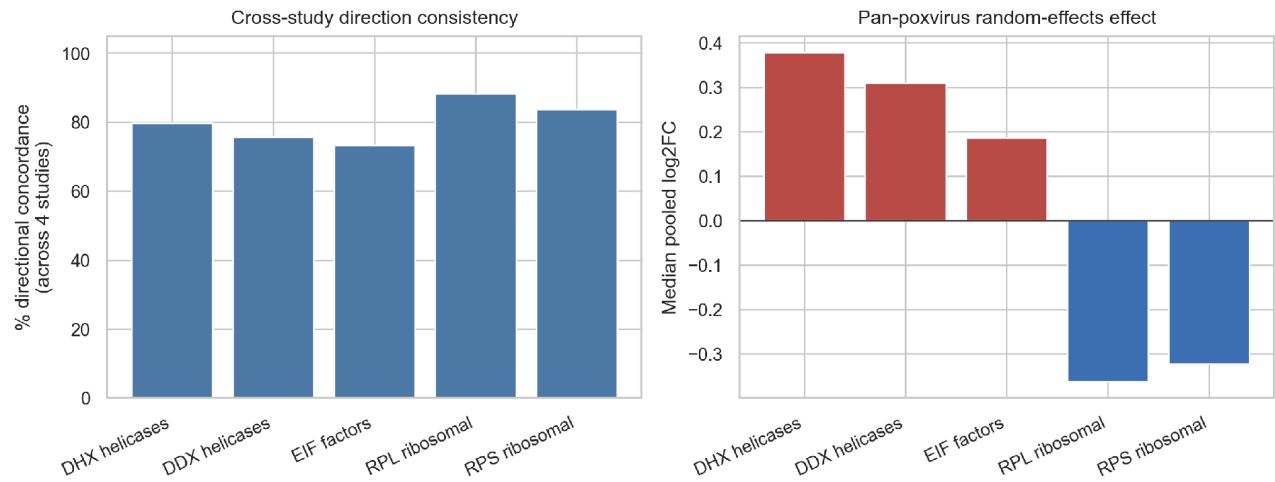


Figure S17. Scope statement and falsifiable prediction.

PoxHostAtlas is computational and hypothesis-generating; it establishes reproducibility of direction across independent studies, not causal mechanism. Prediction: in the next independent poxvirus host-response dataset added, the reproducible down-regulation of cell-cycle and ribosomal-protein programs should replicate and DHX15 should remain among the top-ranked reproducible helicases. Failure of either would falsify the core claim.